NG_distribution_emissions_r4.R

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Contents

`NG_distribution` *Create gridded natural gas distribution methane emissions maps*

Description

'NG_distribution' writes up to 63 netcdf files of gridded methane emissions from natural gas distribution sources, as well as optional visuals

Usage

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```

```
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  County_Tigerlines
)
```

Arguments

`domain` SpatVector polygon outlining the desired output area

`domain_template` SpatRaster providing the desired output grid, including the desired resolution and coordinate reference system

`state_name_list` Character vector listing all states within the desired domain

`input_directory` Character providing the full filepath to save/load raw input data

`output_directory` Character providing the full filepath to save processed data

`inventory_year` Character indicating the desired year of data to use.

`verbose` Logical indicating whether to save additional output. This includes plots of the gridded methane emissions for each fuel-sector-inventory-variation combination as well as 2 summed plots for each inventory-variation combination - one for wood and one for all other sectors.

`ghgrp_facility_info` Data.frame with the GHGRP location data for all years and states. See https://www.epa.gov/enviro/envirofacts-data-service-api

`EIA_file` Character providing the full filepath to the EIA Form 176 - Annual Report of Natural and supplemental Gas Supply and Disposition xlsx file available at https://www.eia.gov/naturalgas/ngqs/#?report=RP4&year1=2020&year2=2020&company=Name. The year can be set to just the desired year. Then all data can be downloaded to an xlsx file using a button to the topright of the data table. If running at the LDC level, the file must be edited such that there is a consistent identifier with other input data (HIFLD, PHMSA, GHGRP). There is an example file in the package's datasets folder that has been successfully used in this code available for reference.

`PHMSA_file` Character providing the full filepath to the PHMSA Gas Distribution Annual Data xlsx file available at https://www.phmsa.dot.gov/data-and-statistics/pipeline/gas-distribution-gas-gathering-gas-transmission-hazardous-liquids. Zip files for groups of years are available via links at the bottom of the page. If running at the LDC level the file must be edited such that there is a consistent identifier with other input data (EIA, HIFLD, GHGRP). There is an example file in the package's datasets folder that has been successfully used in this code available for reference.

`GHGI_file` Character providing the full filepath to the GHGI Annex 3.6 excel file. This data is available at https://www.epa.gov/ghgemissions/natural-gas-and-petroleum-systems-gh for the 2022 GHGI. In the GHGI Annexes, available at https://www.epa.gov/

ghgemissions/inventory-us-greenhouse-gas-emissions-and-sinks-1990-2020, there is a link to the file in Section 3.6: "Methodology for Estimating CH4, CO2, and N2O Emissions from Natural Gas Systems". The excel file has multiple sheets, each of which has a separate layout. There is an example file in the package's datasets folder that has been successfully used in this code available for reference.

`GHGI_MnR` Character or data.frame. Pulled from config file. Either GHGI to indicate the GHGI file should be used to pull emissions and activity data or a data frame providing the needed values.

`GHGI_maintenance` Character or data.frame. Pulled from config file. Either GHGI to indicate the GHGI file should be used to pull emissions and activity data or a data frame providing the needed values.

`GHGI_meters` Character or data.frame. Pulled from config file. Either GHGI to indicate the GHGI file should be used to pull emissions and activity data or a data frame providing the needed values.

`GHGI_services` Character or data.frame. Pulled from config file. Either GHGI to indicate the GHGI file should be used to pull emissions and activity data or a data frame providing the needed values.

`State_Tigerlines` SpatVector. United States Census Bureau county shapefile. Available at https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html.

`NG_distribution_by_LDC` Logical. Pulled from config file. Indicating whether emissions should be calculated at the Local Distribution Company level.

`NG_distribution_by_state` Logical. Pulled from config file. Indicating whether emissions should be calculated at the state level.

`NG_distribution_by_domain` Logical. Pulled from config file. Indicating whether emissions should be calculated at the domain level.

`GHGI_natural_gas_pipeline_emission_factors` Data frame. Pulled from config file. 2 columns by 4 rows. Columns are "Leaks_per_mile" and "Avg_emissions_mol_per_s". Rownames are "Bare_Steel", "Cast_Iron", "Coated_steel", and "Plastic". Default is from Weller et al.

`natural_gas_post_meter_emission_factor` Numeric. Pulled from config file. Emission factor for whole-house residential emissions in mol/s. Default is from Fischer et al.

`Use_ACES` Logical indicating whether or not to use ACES to disaggregate from county-level to pixel level emissions. Either ACES or Vulcan must be used, though both can be.

`Use_Vulcan` Logical indicating whether or not to use Vulcan to disaggregate from county-level to pixel level emissions. Either ACES or Vulcan must be used, though both can be.

`ACES_directory` **Optional** character providing the full path to a folder containing the ACES sectoral CO2 inventories. Must include the residential and commercial sectors. ACES v2.0 is available at https://doi.org/10.3334/ORNLDAAC/1943, though the hourly file should be averaged across hours to create an annually averaged inventory. Code to do this on a linux-based HPC system is available as the script "Annualize_ACES_seawulf.R" and the accompanying batch script "Annualize_ACES.sh". The year closest to "inventory_year" is used, but those further from that year are considered if the closest is unavailable. Only needed if Use_ACES = TRUE. At least one of Use_Vulcan or Use_ACES must be TRUE.

`vulcan_directory`
: **Optional** character providing the full path to a folder containing the Vulcan sectoral CO2 inventories. Must include the residential and commercial sectors. Vulcan v3.0 is available at https://doi.org/10.3334/ORNLDAAC/1741, and the annual mean files should be used. The year closest to "inventory_year" is used. As all years are contained in the same file, it does not search for other years. Only needed if Use_Vulcan = TRUE. At least one of Use_Vulcan or Use_ACES must be TRUE.

`ACES_year`
: **Optional** numeric providing the year of ACES data to use. Only needed if Use_ACES = TRUE. At least one of Use_Vulcan or Use_ACES must be TRUE.

`vulcan_band`
: **Optional** numeric providing the band of Vulcan data to use (1-6 = 2010 - 2015). Only needed if Use_Vulcan = TRUE. At least one of Use_Vulcan or Use_ACES must be TRUE.

`plot_directory`
: **Optional** character providing the full filepath to save figures. Only needed if verbose = TRUE.

`County_Tigerlines`
: **Optional** spatVector. United States Census Bureau county shapefile downloaded in Main. Only needed if verbose = TRUE.

Details

This function calculates and grids methane emissions from natural gas distribution. It uses a Homeland Infrastructure Foundation-Level Data (HIFLD) dataset titled Natural Gas Local Distribution Company Service Territories, Environmental Protection Agency's (EPA) Greenhouse Gas Inventory (GHGI), the EPA Greenhouse Gas Reporting Program (GHGRP), the Pipeline and Hazardous Materials Safety Administration (PHMSA) gas distribution annual report, the Energy Information Administration (EIA) Form 176 - Annual Report of Natural and Supplemental Gas Supply and Disposition, and either the Vulcan or Anthropogenic Carbon Emission System (ACES) CO2 inventory. This can be done at the Local Distribution Company (LDC) level, State level, or Domain level. As several of the input datasets have no common LDC identifier, they must be carefully matched to run this function at the LDC level. If running at the state or domain level, data can be aggregated and applied between datasets without any manual matching needed.

First all of the input data excluding the GHGI are loaded in and filtered to only the states within the domain. The GHGRP data is used to calculate the typical number of stations per mile of pipeline for each LDC. Stations include transmission and distribution transfer stations and metering and regulating stations. This is calculated for all stations combined, as well as separately for above grade and below grade stations. If calculating by LDC, the PHMSA miles of pipeline per LDC is used in the calculation of stations per mile, rather than the GHGRP, though the miles of pipeline for each LDC in GHGRP and PHMSA are compared and if any LDC differs by > 5% an error will be flagged.

The GHGI Annex data is then pulled and includes emission factors and activity data for metering an regulating stations separated by inlet pressure, and just the emission factors for pipeline services separated by pipeline material, meters separated by customer type (residential, commercial, industrial), and different maintenance events (pressure relief valve release, blow-downs, accidental dig-ins).

PHMSA data on miles of pipeline separated by material is then combined with the input "GHGI_natural_gas_pipeline_em which by default are the emission factors from Table 2 of Weller et al. who used research equipped

Google Street View cars to measure > 4000 leaks in the U.S. PHMSA data on pipeline services, separated by pipeline material, are combined with the GHGI emission factors to get emissions for each LDC.

GHGRP data is aggregated to the state level to get the average number of above or below grade stations per mile. If not calculating emission by LDC, this is combined with the PHMSA miles of pipeline to get an estimate of the number of above and below grade stations for each LDC in the PHMSA dataset.

The combined miles of pipeline including services is calculated from the PHMSA and the various input datasets are subset to only the necessary variables, aggregated to the state level and merged. The variables kept differ slightly if calculating by LDC.

If not calculating by LDC, then there is no HIFLD csv file necessary, leaving the PHMSA activity data, the EIA sales data, and the GHGRP activity data.

If calculating by LDC a HIFLD csv and shapefile (containing the same information) are also included. The residuals (LDCs that could not be matched across datasets) are assigned to an "OTHER" LDC that includes all land in the state not already accounted for by an LDC. The number of stations for each LDC is now overwritten to be the values for that specific LDC from the GHGRP. State average stations per mile (calculated from the GHGRP) are then applied only to LDCs that did not report to the GHGRP.

Service and pipeline emissions are then split into residential and commercial fractions. This is calculated for residential as the sum of all emissions * N residential customers / N total residential and commercial customers. The calculation would be equivalent for commercial customers.

Emissions for metering and regulating stations are then calculated as a function of pressure. The number of stations of a grade (above or below) are multiplied with the national fraction of metering and regulating stations of that grade that are a certain pressure window and then multiplied by the emission factor for that pressure window. E.g., N Above grade (by LDC or state) * national type fraction * national type emission factor. These emissions are then split into residential and commercial in the same manner as service and pipeline emissions were.

Meter emissions are calculated separately for each customer type (residential, commercial, industrial) using the corresponding GHGI emission factors. These emissions are then split into residential and commercial, incorporating industrial emissions into residential and commercial proportionally. This was done as the industrial sectors of ACES/Vulcan are dominated by point sources, many of which don't even rely on natural gas. It was chosen to split by emissions rather than number of customers as this was considered more representative (e.g., there may be many residential customers, yet the commercial ones consume more natural gas overall).

Maintenance and upsets are calculated using national GHGI emission factors and split into residential and commercial in the same manner as service and pipeline emissions were.

Post-meter emissions are calculated, strictly for residential, using the volume of gas delivered to residential customers and the natural_gas_post_meter_emission_factor, which is based on Fischer et al. by default. Fischer et al. measured whole-house emissions from 75 homes in California using mass balance.

Finally, ACES and/or Vulcan residential and commercial gridded CO_2 emission maps are loaded in. The emissions for each subsector are then distributed using the ACES or Vulcan CO_2 inventory. This can be done at the LDC, state, or domain level. Emissions at this point are at the state level if not calculating by LDC and can then be aggregated to the domain level. Otherwise emissions are at the LDC level and can be aggregated to the state or domain level. As such, producing output at the state and LDC level will result in slightly different output than running only at the state level.

So, to summarize this relatively complex sector, PHMSA data on miles of pipeline by type and number of services by type of pipeline is combined with emission factors to calculate emissions. GHGRP state average numbers of M&R facilities per mile are also combined with an emission factor, broken down by pressure. If calculating byLDC and GHGRP data exists for an LDC, the reported counts are used instead. These emissions are then distributed to the appropriate LDC territories using HIFLD shapefiles, or aggregated at the state level. To further disaggregate the emissions, they are broken into residential and commercial portions using the fraction of customers as reported to the EIA. They are then distributed to the pixel scale using the residential and commercial sectors of the CO2 inventories Vulcan/ACES. As such, GHGRP emissions are not directly used at all, PHMSA is the source of most activity data with GHGRP providing the numbers of M&R facilities, HIFLD solely provides shapefiles for each LDC if operating by LDC, and EIA provides a breakdown of residential vs commercial customers.

GHGRP data is available starting in 2010 and generally is about 2 years behind present day, the GHGI is available starting in 1990 and is updated approximately in sync with the GHGRP. The HIFLD dataset is updated infrequently and is available for 2019 and 2017. The EIA form is annually reported and is available starting in 1997 and is generally updated in September to the previous year (i.e., Sept 2024 adds 2023 data). The PHMSA data is annual and is available starting in 1970 and is available up to the most recent year. The GHGRP includes only facilities that emit at least 25,000 metric tons of carbon dioxide equivalent while the GHGI is intended to capture all national emissions. All other datasets are meant to be inclusive of national facilities. All data is annual. The GHGI is national totals while all other datasets are at the facility scale.

GHGRP data and the HIFLD shapefile will be automatically downloaded.

The GHGRP is available at `https://ghgdata.epa.gov/ghgp/main.do`. The GHGI is available at `https://www.epa.gov/ghgemissions/inventory-us-greenhouse-gas-emissions-and-sinks`. For individual LDC metering and regulating station counts, among other variables, one must filter to the natural gas local distribution companies sector, select an individual facility and select "View reported data". The HIFLD dataset is available at `https://hifld-geoplatform.hub.arcgis.com/datasets/geoplatform::natural-gas-service-territories/about` and can be donwloaded as both a shapefile or a csv. EIA form 176 is available at `https://www.eia.gov/naturalgas/ngqs/#?report=RP4&year1=2020&year2=2020&company=Name` and can be downloaded as an excel file. The PHMSA Gas Distribution Annual Data can be download at `https://www.phmsa.dot.gov/data-and-statistics/pipeline/gas-distribution-gas-gathering-gas-transmission-hazardous-liqu` as a zip file with an excel file for each year. ACES is available at `https://doi.org/10.3334/ORNLDAAC/1943` and Vulcan is available at `https://doi.org/10.3334/ORNLDAAC/1741`.

See references Weller et al., Fischer et al., Vulcan and, ACES

Value

Nothing is returned from the function, but the main outputs are up to 63 netcdf files of the methane emissions from natural gas distribution. They are titled as "NG_dist_type_sector_variation.nc" where type is upset, serv (services), post_meter, MnR (metering and regulating stations), and mains; sector is abbreviated as res (residential) or com (commercial); and variation is byLDC, bystate, or bydomain.

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Examples

```
library(terra)
grid_bbox=cbind(c(-76.65,-73.65),c(38.97,40.97))
```

```
grid_res=0.01
grid_crs="epsg:4326"
grid <- rast(nrows=diff(range(grid_bbox[,2]))/grid_res,
             ncols=diff(range(grid_bbox[,1]))/grid_res, xmin=min(grid_bbox[,1]),
             xmax=max(grid_bbox[,1]), ymin=min(grid_bbox[,2]), ymax=max(grid_bbox[,2]),
             crs=grid_crs)
grid_vect <- as.polygons(ext(grid),crs=grid_crs)

GHGI_EF_dataframe <- data.frame("Leaks_per_mile"=
                                  c(0.51,1,0.61,0.43),
                                "Avg_emissions_mol_per_s"=
                         c(2.24,1.72,2,2.03)/(16.043*60)) #converting from g/min to mol/s
rownames(GHGI_EF_dataframe) <- c("Bare_Steel",
                                 "Cast_Iron",
                                 "Coated_steel",
                                 "Plastic")

NG_distribution(domain=grid_vect,
                domain_template=grid,
                state_name_list=c("DE","MD","NJ","NY","PA"),
                input_directory="~/../Desktop/in/",
                output_directory="~/../Desktop/out/",
                inventory_year=2018,
                verbose=TRUE,
                ghgrp_facility_info="~/../Desktop/in/GHGRP/facility_info.csv",
               EIA_file = "~/../Desktop/in/176 Type of Operations and Sector Items.xlsx",
          PHMSA_file = "~/../Desktop/in/annual_gas_distribution_2010_present/annual_gas_distribution_2019.x
          GHGI_file = "~/../Desktop/in/2022_ghgi_natural_gas_systems_annex36_tables.xlsx",
                GHGI_MnR="GHGI",
                GHGI_maintenance="GHGI",
                GHGI_meters="GHGI",
                GHGI_services="GHGI",
          State_Tigerlines=vect("~/../Desktop/in/State_Tigerlines/tl_2018_us_state.shp"),
                NG_distribution_by_LDC = FALSE,
                NG_distribution_by_state = TRUE,
                NG_distribution_by_domain = TRUE,
                GHGI_natural_gas_pipeline_emission_factors=GHGI_EF_dataframe,
           natural_gas_post_meter_emission_factor=7850/401*0.005/(16.043*60*60*24*365),
                Use_ACES=TRUE,
                Use_Vulcan=TRUE,
                ACES_directory="~/../Desktop/in/Inventories/ACES_v2.0",
                vulcan_directory="~/../Desktop/in/Inventories/Vulcan_v3.0",
                ACES_year=2017,
                vulcan_band=6,
          County_Tigerlines=vect("~/../Desktop/in/County_Tigerlines/tl_2018_us_county.shp"),
                plot_directory="~/../Desktop/plots/")
```

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